Acyclicity and simultaneous protection of individual core maxima

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Abstract

A stable allocation need not protect each participant relative to that participant's most favorable stable payoff. We characterize when sector interactions permit a uniform simultaneous guarantee of these individual core maxima. Fix a connected graph on $K \geq 2$ equal-sized sectors. Full teams contain one player from every sector, their values sum nonnegative pair values on the allowed edges, and only full teams produce. Coalition worth is the maximum value of an integral matching. Among markets with positive worth and nonempty core, both full leximin and the best choice of any core allocation have sharp size-independent guarantee $1/K$ on trees and zero on graphs containing a cycle. The tree proof describes the entire core by cut coordinates and uses clipping at every lexicographic level. For a triangular family with pair values in $\{0, 1, 2\}$, the entire core is a simplex and the optimal simultaneous fraction is exactly $1/(2q + 3)$. Selective subdivision transfers vanishing protection to every cyclic allowed graph while preserving a positive core allocation. The cyclic obstruction therefore concerns what stability can achieve, not merely the choice of a fairness rule. The classification is conditional on complete-team production and permits zero-valued allowed edges.

Keywords: assignment game; core; leximin; individual ideal; simultaneous protection; sector graph.
JEL classification: C71; C78.

1 Introduction

When production requires partners from several sectors, coalitional stability and individual protection address different questions. The core rules out a group earning more by producing separately. It does not ensure that a player receives a substantial share of the most that player could obtain in another stable agreement. This distinction matters when the core is used as a feasible set for bargaining: a payoff may be stable yet small relative to the same participant's attainable stable payoff.

For an entire core $\mathcal{C}$, write $H_i = \max_{y \in \mathcal{C}} y_i$. We ask whether one stable allocation can guarantee $p_i \geq \lambda H_i$ simultaneously for all players, with λ depending only on the number and pattern of sector interactions, not the number of players. We also ask whether full leximin, which successively maximizes the smallest nominal payoffs, attains such a uniform guarantee. The benchmark H_i is an attainable core payoff, not an outside option or a claim that all maxima can be attained together. We keep this benchmark fixed when comparing allocations within a market.

Our model describes a production project that needs one participant from every sector. Pair values describe contributions to a complete project's surplus; an edge does not by itself authorize a productive bilateral coalition. The same sharp graph boundary governs both questions. Trees guarantee $1/K$, and full leximin attains that bound for everyone at once. Every cyclic allowed graph instead admits growing markets with vanishing best simultaneous protection, even though each market has a strictly positive core allocation. Thus the negative result cannot be repaired by selecting a different point of the same core. The poorly protected player may depend on that point; each individual maximum is nevertheless attained somewhere in the core.

The proof has three parts. A cut-coordinate description of the entire tree core yields a clipping operation that improves any violation of the $1/K$ floor. A fixed-alphabet triangular construction has an explicit simplex core and exact optimal fraction $1/(2q+3)$. Finally, selective subdivision transfers a unit partial-matching shortfall to all cyclic allowed graphs. The latter two steps keep the number of sectors fixed as market size grows. The results measure distributional possibilities under stability, not a loss in aggregate surplus, typical-market performance, or a change in productive efficiency.

Related work. The multisided assignment model and its core have a substantial literature, including Quint (1991) and Lucas (1995). The strict supplier–firm–buyer model is associated with Stuart (1997); Atay and Solymosi (2018), Definition 3 and equations (1)–(2), provide a later full formulation with signed adjacent-pair values and zero-truncated triple totals. Our nonnegative three-sector path is a subclass. We claim neither the model nor chain-core nonemptiness as new.

The closest graph comparison is Atay and Núñez (2019). Their Theorem 1 expresses tree cores through bilateral cores, Corollary 1 gives a tree/cycle strong-balancedness boundary, and Proposition 2 gives simultaneous maxima within a sector. Their connected partial-sector teams also produce value. Our core instead enforces strict full-team production and the cyclic theorem is conditional on its nonemptiness. Thus neither their core decomposition nor their empty-core cyclic examples establish the protection classification here. The elementary comparison below shows explicitly how partial-team production changes the individual maxima.

Simultaneous ideal protection has a direct bilateral precedent: the half-optimal core of Núñez and Rafels requires at least half of each player's maximum core payoff. The original working text is Núñez and Rafels, *Bargained stable allocations in assignment markets*, Barcelona Economics Working Paper 153, December 28, 2004, Definition 6, printed p.19, with sharpness in Example 7, p.20 (`https://www.bse.eu/sites/default/files/working_paper_pdfs/153.pdf`). The bilateral bargaining midpoint is also studied by Núñez and Rafels (2008). Neither the benchmark nor the bilateral half guarantee is proposed as a new fairness concept.

Gangam et al. (2026), p.1621, give a unit-capacity path whose equitable payoff is $1/r$ on each of its r edges. Its entire core is the unit simplex; therefore each ideal equals one and even the best simultaneous fraction is $1/r$. This last conclusion is an immediate inference from that example. Its productive coalition size grows with r, whereas our obstruction already uses strict triples and pair values in $\{0,1,2\}$. The distinction is fixed production-team size within a restricted class, not the first occurrence of vanishing individual protection in cooperative games.

Extreme-core descriptions for assortative multisided markets (Martínez-de-Albéniz et al., 2020) and simultaneous extrema under supermodularity and local monotonicity (Martínez-de-Albéniz et al., 2026) impose different restrictions. A general 2023 core-redescription preprint of the same authors also retains original inequalities outside a central strip: Martínez-de-Albéniz, Rafels and Ybern, *Extreme core points in multisided assignment problems*, 2023 preprint, Theorem 2 and Theorems 3–4 (`https://doi.org/10.2139/ssrn.4369524`). That general representation may apply to our

inputs; non-supermodularity does not exclude it. It does not itself give the quantitative protection classification proved here. We do not assert an exclusion under every relabeling or assume an arbitrary core representation preserves the payoff labels, selector and pair-value restrictions relevant to this question.

Matching duality, individual ideals, leximin, convex averaging and closure of difference constraints under minimum are established ingredients. The proposed contribution is their specific constructive use to classify protection on the entire strict full-team core, with a branching-tree argument and a bounded-alphabet obstruction on every cyclic allowed graph. Full-original comparisons with Quint, Lucas and Stuart remain incomplete under freely available access; their basic attribution is retained, without assuming uninspected detailed statements or claiming exhaustive priority clearance. No theorem below relies on such an uninspected statement.

2 Model and main classification

Fix a connected simple graph $G = (V, E)$ with $K = |V| \geq 2$. Each sector $v \in V$ contains players (v, j), $1 \leq j \leq m$, for a common finite $m \geq 1$. An allowed edge uv carries a nonnegative matrix A_{uv}; zero matrices are permitted. All full teams $t = (j_v)_{v \in V}$ are available, with worth

$$w(t) = \sum_{uv \in E} A_{uv}(j_u, j_v).$$

Only full teams produce value. For any coalition S, its worth $v(S)$ is the maximum sum of worths of player-disjoint full teams contained in S. In particular every singleton has worth zero. Set $T = v(N) > 0$ and require that the entire core $\mathcal{C}$ be nonempty. Equivalently,

$$\mathcal{C} = \{p \geq 0 : \sum_i p_i = T,\ \sum_{i \in t} p_i \geq w(t) \text{ for every full team } t\}. \tag{1}$$

Indeed these conditions imply every coalition inequality by summing over its integral matching and retaining nonnegative unmatched payoffs.

The compact convex set $\mathcal{C}$ has coordinate maxima H_i. Let x be its full ascending-sorted leximin allocation; all labeled coordinates, including permanent zeros, remain in the optimization. Existence follows by successive maximization of order statistics on compact optimal sets. Uniqueness follows by averaging two distinct purported optima: at their first differing payoff level the average reduces its multiplicity without changing lower levels. Define

$$c(G) = \inf_M \min_{i : H_i > 0} \frac{x_i}{H_i}, \qquad \rho(M) = \max_{p \in \mathcal{C}} \min_{i : H_i > 0} \frac{p_i}{H_i}, \qquad R(G) = \inf_M \rho(M),$$

where M ranges over the stated markets on fixed G, allowing m to grow. Thus $c(G) \leq R(G)$. The ratios are formed only when $H_i > 0$; other coordinates are not deleted from the core or leximin problem.

Theorem 2.1. *For every fixed connected allowed graph on $K \geq 2$ sectors,*

$$c(G) = R(G) = \begin{cases} 1/K, & G \text{ is a tree,} \\ 0, & G \text{ contains a cycle.} \end{cases}$$

For every cyclic G and integer $q \geq 1$, the upper witnesses have $3q + 1$ players per sector, pair entries in $\{0, 1, 2\}$, and a strictly positive core allocation. One target has ideal 1 and full leximin payoff $1/[K(q + 1)]$, while $\rho(M) \leq 1/(2q + 2)$. For the source triangle, the exact stronger formula is $\rho(M_q) = 1/(2q + 3)$.

Every fixed finite market still has $\rho(M) > 0$: average one core maximizer for each of its n_+ positive ideals to obtain $p_i \geq H_i/n_+$. The zero in the theorem is an infimum over sizes, not a finite-market zero. Also, the theorem concerns allowed edges, not strictly positive matrices on every allowed edge. A single-sector edge-sum model has no positive-worth instance and is excluded.

Why complete-team production matters. Consider three sectors with one player each on the path 1–2–3, with both edge values equal to one. In the present model only the triple produces, so $T = 2$ and the core is $\{p \geq 0 : p_1 + p_2 + p_3 = 2\}$. All three ideals equal two and the best simultaneous fraction is $1/3$. If adjacent pairs can also produce their edge values, the core additionally requires $p_1 + p_2 \geq 1$ and $p_2 + p_3 \geq 1$. Its ideals become $(1, 2, 1)$; the point $(1/2, 1, 1/2)$ achieves the optimal fraction $1/2$, since the ideals sum to four while the total remains two. Thus admitting partial teams changes the benchmark as well as the feasible set. A tree theorem for that model does not transfer the same relative-payoff guarantee to the present core.

3 Trees: entire-core coordinates and clipping

Root the tree at an arbitrary sector r. Choose an optimal perfect matching on each edge and propagate labels from the root so that each is diagonal. There is no conflicting propagation on a tree. Write $d_{ab}(j) = A_{ab}(j, j)$ for an oriented parent–child edge $a \to b$. Any full-team matching projects to edge matchings; conversely the diagonal full-team matching attains the sum of these edge optima. Hence $T = \sum_j \sum_{ab \in E} d_{ab}(j)$. Nonnegative weights and equal sizes permit completion of partial matchings without loss. Summing optimal nonnegative bilateral dual covers over incident edges gives a full-team cover of total T, so tree cores are nonempty. Every diagonal team is tight throughout the core.

For a nonroot vertex b, let S_b be its descendant subtree, $s_b = |S_b|$, and define

$$U_b^p(j) = \sum_{v \in S_b} p_v(j) - \sum_{e \text{ internal to } S_b} d_e(j), \qquad U_r^p(j) = 0.$$

Lemma 3.1. *The entire core is in bijection with the cut vectors satisfying*

$$U_b(i) - U_b(j) \geq A_{ab}(j, i) - d_{ab}(j), \tag{2}$$

$$p_v(j) = U_v(j) - \sum_{b \text{ child of } v} U_b(j) + \sum_{b \text{ child of } v} d_{vb}(j) \geq 0. \tag{3}$$

Proof. For necessity, take label i on S_b and label j on its complement in a full team, and subtract the tight diagonal team at j. The single cut edge is ab; internal diagonal terms cancel to give (2). Subtracting child-subtree sums gives (3); at the root use diagonal tightness. Conversely reconstruction telescopes at each fixed label to the diagonal team value and therefore gives total T. At arbitrary labels,

$$\sum_v p_v(i_v) = \sum_{a \to b} [U_b(i_b) - U_b(i_a) + d_{ab}(i_a)] \geq \sum_{a \to b} A_{ab}(i_a, i_b).$$

Thus every full team is covered. Nonnegativity and (1) imply core membership. Descendant sums recover the cut coordinates, proving bijectivity. □

Lemma 3.2 (Clipping)**.** *For $x, y \in \mathcal{C}$ and $\beta \geq 0$, set $Z_b(j) = \min\{U_b^x(j), U_b^y(j) + s_b\beta\}$ for $b \neq r$, and $Z_r = 0$. The reconstructed z belongs to $\mathcal{C}$ and satisfies*

$$z_v(j) \geq \min\{x_v(j), y_v(j) + \beta\} \quad (v \neq r), \qquad z_r(j) \geq \max\{x_r(j), y_r(j) - (K-1)\beta\}.$$

Proof. Difference inequalities $u(i) - u(j) \geq c_{ij}$ are closed under uniform translation and coordinate-wise minimum: both candidate values at i are at least their respective value at j plus c_{ij}. Thus Z satisfies all structural inequalities. At a nonroot vertex, if its selected cut is the x cut, each child cut is at most its x cut and reconstruction gives $z_v \geq x_v$. If it is the shifted y cut, use $s_v = 1 + \sum_b s_b$ to obtain $z_v \geq y_v + \beta$. These cases include ties. At the root, upper-bound each child cut first by its x value and then by its shifted y value; the child subtrees have total size $K - 1$. This gives both root inequalities. All reconstructed payoffs are nonnegative; Lemma 3.1 finishes the proof. □

Fix a target at the root sector and put $\alpha = x_i$ for full leximin x. If $H_i > K\alpha$, choose an ideal-attaining y and $\alpha < \beta < H_i/K$. Clipping gives $z_i \geq H_i - (K-1)\beta > \beta > \alpha$. Every old coordinate at most α weakly increases, and every old coordinate above α remains strictly above it. Thus, for each $\gamma \leq \alpha$, the number of coordinates below γ cannot increase, and the set of coordinates at most α strictly shrinks because the target leaves it. Decreases above α do not affect this comparison. If a lower level increases, its first changed multiplicity decreases; otherwise the multiplicity of α decreases. Either case strictly improves the sorted vector, a contradiction. Consequently $x_i \geq H_i/K$, even when $x_i = 0$. Re-rooting the tree and relabeling its optimal matching change only the coordinates used in the proof, not the chosen leximin allocation x. The target was arbitrary, so this same x protects every player and $\rho(M) \geq 1/K$ as well.

For sharpness take $m = 1$ and edge values totaling one. The sole full team has worth one and every proper coalition has worth zero. Its core is the K-coordinate unit simplex; $H_i = 1$, leximin is uniform, and $\rho = 1/K$. This proves both tree equalities. It also gives the sharp distance bound $\|x - y\|_1/T \leq 2(1 - 1/K)$ for any tree-core point y, since $\min(x_i, y_i) \geq y_i/K$ and both totals are T.

4 A cyclic family with vanishing protection

Fix an integer $q \geq 1$. Use the common team-label set

$$\mathcal{T} = \{L_0\} \cup \{O_s, L_s, R_s : 1 \leq s \leq q\}.$$

For every label t, create players a_t, b_t, c_t in the respective sectors. Each sector has $3q + 1$ players. Define $w(a_t, b_u, c_v) = A_{tu} + B_{tv} + C_{uv}$ by the nonzero entries in Table 1; all unspecified pair entries are zero.

Table 1: Nonzero pair values in the triangular market

Pair entries	Range	Value
C_{L_s,L_s}	$0 \leq s \leq q$	1
C_{R_s,R_s}	$1 \leq s \leq q$	1
$A_{O_s,O_s},\ B_{O_s,O_s}$	$1 \leq s \leq q$	2
$A_{O_s,R_s},\ B_{O_s,L_s}$	$1 \leq s \leq q$	2
$C_{R_s,L_s},\ C_{L_{s-1},O_s}$	$1 \leq s \leq q$	1

There are $8q + 1$ nonzero entries. The diagonal perfect matching consists of (a_t, b_t, c_t) for all $t \in \mathcal{T}$. Each L or R triple has value one and each O triple has value four, so this matching is worth $6q + 1$.

4.1 Two explicit full-core covers

Put $\tau = 1/[3(q+1)]$. Define allocations u and y in (a, b, c) order:

team	u	y
L_s	$(\tau,\ 1-(3s+2)\tau,\ (3s+1)\tau)$	$(0,0,1)$
R_s	$(\tau,\ 1-2\tau,\ \tau)$	$(0,1,0)$
O_s	$(4-(3s-1)\tau,\ \tau,\ (3s-2)\tau)$	$(3,0,1)$.

(4)

The L row uses $0 \le s \le q$; the other rows use $1 \le s \le q$. Every coordinate of u is at least $\tau > 0$, and y is nonnegative. The smallest L-sector b coordinate is attained at $s = q$ and equals τ; the smallest O-sector c coordinate is attained at $s = 1$ and also equals τ. Each diagonal triple receives its value at both points, so both allocations have total $6q + 1$.

We check every triple cover, including all mixed-team triples. If its a player has an L or R label, its A and B entries are zero. Table 2 lists the possible positive C pairs and the associated two-player sums.

Table 2: Coverage when the first-sector player has a weak label

(b, c) labels	$u_b + u_c$	$y_b + y_c$
(L_s, L_s)	$1-\tau$	1
(R_s, R_s)	$1-\tau$	1
(R_s, L_s)	$1+(3s-1)\tau$	2
(L_{s-1}, O_s)	$1-\tau$	1

All four pair values are one. Adding $u_a = \tau$ or $y_a = 0$ covers the triple. Zero C entries are covered by nonnegativity.

Now suppose its a player is a_{O_s}. If at most one of its A and B entries is positive, the triple value is at most three, covered by $u_{a_{O_s}} \ge 3 + 4\tau$ and $y_{a_{O_s}} = 3$. If both entries are positive, exactly the four cases in Table 3 remain.

Table 3: Coverage when both adjacent pair values are positive

(b, c) labels	w	$u_a + u_b + u_c$	$y_a + y_b + y_c$
(O_s, O_s)	4	4	4
(O_s, L_s)	4	$4+3\tau$	4
(R_s, O_s)	4	$5-3\tau$	5
(R_s, L_s)	5	5	5

Since $\tau \le 1/6$, all are covers. The cases also show that every triple value is at most five.

The matching gives worth at least $6q + 1$, while either nonnegative cover bounds every matching by its total $6q + 1$. Hence the true integral worth is $T = 6q + 1$ and both u, y belong to the entire core. No fractional matching value is substituted for coalition worth.

4.2 A partial matching determines the target

For each $s = 1, \ldots, q$, take the triples

$$N_s = (a_{O_s}, b_{R_s}, c_{L_s}), \qquad E_s = (a_{L_{s-1}}, b_{L_{s-1}}, c_{O_s}).$$

Their values are five and one, respectively. They form a player-disjoint partial matching: the used a labels are the O labels and $L_0, \dots, L_{q-1}$; the used b labels are the R labels and the same L labels; the used c labels are $L_1, \dots, L_q$ and the O labels. Its total is $6q$. Its unmatched players are exactly

$$U = \{c_{L_0}, a_{L_q}, b_{L_q}\} \cup \{a_{R_s}, b_{O_s}, c_{R_s} : 1 \le s \le q\}, \qquad |U| = 3q + 3. \tag{5}$$

Summing the partial-matching inequalities and subtracting from efficiency gives, at every core point p,

$$\sum_{a \in U} p_a \le 1. \tag{6}$$

Thus a common payoff floor cannot exceed $1/(3q+3) = \tau$. The point u attains this floor, so the first global leximin level is exactly τ. At every point of the first-stage optimal set, all coordinates are at least τ, and $|U|\tau = 1$. Inequality (6) therefore fixes every coordinate of U at τ on that entire set. In particular, the full leximin target c_{L_0} is fixed at τ through every later refinement. We do not need, and do not assert, that u is the complete final leximin vector.

The diagonal L_0 triple is tight at every core point and has value one. Nonnegativity implies $H_{c_{L_0}} \le 1$, while the cover y attains one at that coordinate. Consequently

$$H_{c_{L_0}} = 1, \qquad x_{c_{L_0}} = \frac{1}{3(q+1)}.$$

This proves the triangular nominal-leximin conclusion. The next section characterizes the best simultaneous protection over the entire core.

5 The entire triangular core and optimal simultaneous protection

5.1 A simplex representation

All weak a-players, meaning those at L or R labels, have identical productivity. Replace the weak a-player in a tight diagonal triple by any other weak a-player. The resulting inequality compares their payoffs; reversing the replacement proves equality. Denote their common payoff by $\alpha \ge 0$. Write

$$l_s = p_{c_{L_s}} \ (0 \le s \le q), \qquad r_s = p_{c_{R_s}},\ d_s = p_{b_{O_s}},\ o_s = p_{c_{O_s}} \ (1 \le s \le q).$$

Diagonal tightness gives

$$p_{b_{L_s}} = 1 - \alpha - l_s, \quad p_{b_{R_s}} = 1 - \alpha - r_s, \quad p_{a_{O_s}} = 4 - d_s - o_s.$$

The critical triples E_s, N_s respectively give $o_s \ge l_{s-1}$ and $l_s \ge \alpha + r_s + d_s + o_s$. Put $e_s = o_s - l_{s-1}$, $f_s = l_s - \alpha - r_s - d_s - o_s$, and $h = p_{b_{L_q}}$. These variables satisfy

$$(q+1)\alpha + l_0 + \sum_{s=1}^{q} (d_s + r_s + e_s + f_s) + h = 1, \qquad \alpha, l_0, d_s, r_s, e_s, f_s, h \ge 0. \tag{7}$$

Proposition 5.1. *Equation* (7)*, with recursive reconstruction*

$$o_s = l_{s-1} + e_s, \qquad l_s = l_{s-1} + \alpha + d_s + r_s + e_s + f_s,$$

characterizes the entire core of M_q. In particular the core is affinely isomorphic to a simplex of dimension $4q + 2$.

Proof. Necessity follows from the displayed tight teams and inequalities. For sufficiency, l_s is nondecreasing and $l_q = 1-\alpha-h \le 1-\alpha$, proving nonnegativity of all b_L payoffs. Also $r_s \le l_s - \alpha \le 1-2\alpha$, giving nonnegativity of b_R. Since $d_s + o_s \le l_s - \alpha - r_s \le 1$, every strong a-payoff is at least three. The other coordinates are visibly nonnegative and diagonal sums give efficiency.

For a weak a-player, the positive C edges of types $(L_s, L_s), (R_s, R_s), (R_s, L_s), (L_{s-1}, O_s)$ have total triple payoffs

$$1, \quad 1, \quad 1 + l_s - r_s, \quad 1 + o_s - l_{s-1},$$

respectively, all at least their worth one. If a strong a_{O_s} has at most one positive adjacent A or B entry, its own payoff of at least three covers the triple's worth of at most three. When both are positive, the four (b, c) label pairs $(O_s, O_s), (O_s, L_s), (R_s, O_s), (R_s, L_s)$ have worths $4, 4, 4, 5$ and respective payoff sums

$$4, \quad 4 + l_s - o_s, \quad 5 - \alpha - r_s - d_s, \quad 5 + l_s - \alpha - r_s - d_s - o_s.$$

They cover their worths because $l_s \ge o_s$, $\alpha + r_s + d_s \le l_s \le 1$, and $f_s \ge 0$. Zero-edge cases are covered by nonnegativity. This exhausts all full triples and proves core membership by (1).

The parameters are recovered from the payoffs and differences already specified, so reconstruction is injective. After replacing α by $(q+1)\alpha$, equation (7) has $4q+3$ nonnegative barycentric coordinates summing to one, hence dimension $4q+2$. □

5.2 All ideals and the exact best fraction

The simplex representation yields

$$H_{a_{L_s}} = H_{a_{R_s}} = \frac{1}{q+1}, \qquad H_{a_{O_s}} = 4, \qquad H_{b_t} = H_{c_t} = 1 \quad \text{for every label } t. \tag{8}$$

For the weak a upper bounds use (7); for strong a use diagonal tightness. The reconstructed formulas bound each b, c coordinate by one: in particular $o_s, d_s, r_s \le l_s \le 1$. Attainment uses separate simplex vertices. Set only $\alpha = 1/(q+1)$ for weak a ideals; only $h = 1$ for all strong a and weak b ideals; only $d_s = 1$ for b_{O_s}; only $r_s = 1$ for c_{R_s}; and only $l_0 = 1$ for all c_L, c_O. No simultaneous attainment is assumed.

For any partial matching P in any admissible market, let U be its unmatched players and $\delta = T - w(P)$. Every core point satisfies $\sum_{i \in U} p_i \le \delta$. Consequently

$$\rho(M) \le \frac{\delta}{\sum_{i \in U} H_i} \qquad \text{if } \sum_{i \in U} H_i > 0. \tag{9}$$

This follows directly by summing the desired floors over U; even lower bounds on those ideals suffice for a weaker upper certificate.

For the triangular critical matching, $\delta = 1$ and the unmatched set in (5) has $q+1$ weak a-players, each of ideal $1/(q+1)$, and $2q+2$ players in sectors b, c, each of ideal one. Thus $\sum_U H_i = 2q+3$ and $\rho(M_q) \le 1/(2q+3)$. For equality put $t = 1/(2q+3)$ and choose

$$\alpha = \frac{t}{q+1}, \quad l_0 = h = t, \quad d_s = r_s = t, \quad e_s = f_s = 0.$$

Equation (7) holds. Every b, c payoff is at least t: $l_s, o_s \ge t$, $d_s = r_s = t$, $b_{L_s} \ge h = t$, and $b_{R_s} = 1 - \alpha - t \ge t$. Weak a-payoffs equal tH_i, and strong a-payoffs are at least $3 \ge 4t = tH_i$. Hence

$$\rho(M_q) = \frac{1}{2q+3}.$$

Already $q = 1$ gives a twelve-player market with best fraction $1/5 < 1/3$. The optimizing point is strictly positive. A lottery over core allocations cannot improve the simultaneous guarantee in expected payoffs: its mean remains in the convex core. This observation makes no claim about other criteria for evaluating lotteries.

6 Selective subdivision and every cyclic allowed graph

6.1 A transfer lemma with explicit hypotheses

Take a strict k-sector market with an optimal perfect matching relabeled diagonally, worth T. Suppose edge uv has matrix $B(i, j)$, with i at u. Choose $D_j \geq \max_i B(i, j)$. Insert a sector d, replacing uv by ud with matrix B and dv with diagonal entries D_j and off-diagonal zeros. For an old team t using u-label i and v-label j, and a new label h, its new worth is

$$w_1(t, h) = w_0(t) - B(i, j) + B(i, h) + D_h \mathbf{1}_{h=j}.$$

For every old core point z, the unshifted lift (z, D) covers every new team. If $h = j$ this follows from the old inequality with the bonus added. If $h \neq j$, use $D_h \geq B(i, h)$ and $B(i, j) \geq 0$. The extended diagonal matching and this cover have the same total $T' = T + \sum_j D_j$, proving integral optimality and lifted core membership. In particular all new diagonal teams are tight throughout the new core.

If z has floor τ, let $t = k\tau/(k + 1)$. Subtract t/k from every old coordinate and give new-sector label j payoff $D_j + t$. Every full-team sum is unchanged. Equal sector sizes make the total change zero, and all coordinates are at least t, so this shifted lift also lies in the core.

Now let an old partial matching P have $r < m$ teams and worth $T - \delta$. Extend each team using new label equal to its v-label. This remains disjoint. If P uses every v-label with positive D_j, it collects all bonuses and keeps shortfall δ. The new unmatched set has $(k + 1)(m - r)$ players. If the old floor witness has $\tau = \delta/[k(m - r)]$, the shifted witness attains the new floor $\delta/[(k + 1)(m - r)]$. The unmatched budget proves that every unmatched coordinate is fixed to that value on the entire first-stage optimal set.

An old unmatched target keeps ideal H if its diagonal team has worth H, its ideal is attained at an old core point, and the bonus at its diagonal label is zero. The tight new diagonal gives the upper bound; the unshifted lift of the attaining point gives the lower bound. More generally every old ideal has at least its previous value by unshifted lifting. No equality of entire cores or preservation of all ideals is needed.

6.2 Applying the lemma repeatedly

Start with the triangle's edge B from a to c. Its positive columns are $\mathcal{H} = \{O_s, L_s : 1 \leq s \leq q\}$ and their maxima are two. Set $D_j = 2$ on $\mathcal{H}$ and zero otherwise. The critical matching uses exactly these labels in sector c, so it collects every bonus. Also $D_{L_0} = 0$, preserving the target ideal one. Each subdivision adds $4q$ to the total worth and keeps the unit shortfall.

Subdivide the edge still carrying B, oriented from a to its current neighbor, repeatedly. The matching labels at that neighbor remain exactly $\mathcal{H}$, and all target diagonal labels remain L_0. Thus every transfer hypothesis propagates. After $\ell - 3$ steps the sector support is an ℓ-cycle, with

$$T_\ell = 6q + 1 + 4q(\ell - 3), \quad w(P) = T_\ell - 1, \quad |U| = \ell(q + 1), \quad x_{c_{L_0}} = \frac{1}{\ell(q + 1)}, \quad H_{c_{L_0}} = 1.$$

All pair entries remain in $\{0, 1, 2\}$ and the shifted floor witness is strictly positive.

Adding a zero-interaction sector leaves full-team worths unchanged. An old core point lifts by zero; a floor witness is shifted exactly as above with $D = 0$. Extend the critical matching by any r distinct new labels. The same unit budget and first-stage pinning arguments apply; the target's diagonal worth and attainable ideal stay one.

Choose a simple cycle in any fixed cyclic allowed graph G. Embed the cycle construction there, add the other sectors with zero interactions, and put zero matrices on all other allowed edges, including chords. This is allowed by the model and gives the stated leximin target $1/[K(q+1)]$ with ideal one.

For simultaneous protection retain the $2q+2$ original b, c players in U. Their old ideal-one witnesses lift without changing those payoffs; they remain unmatched. Thus their enlarged-core ideals are at least one, and (9) gives $\rho(M_{G,q}) \leq 1/(2q+2)$. We do not assert equality for longer cycles. Letting q grow gives $R(G) = 0$, while nonnegativity and $c(G) \leq R(G)$ give $c(G) = 0$. Together with the tree proof, this proves Theorem 2.1.

7 Economic scope and reproducibility

The classification identifies when stability itself permits a guarantee relative to every participant's individually most favorable stable agreement. For a tree, a single familiar fairness selection meets the sharp guarantee. For a cyclic graph, the obstruction applies to the best possible selection. Every finite market still has a positive optimal fraction; the zero guarantee is a worst-case statement as the number of players per sector grows. The strictly positive witness allocations separate this conclusion from an empty core or a player who can never be paid positively.

The theorem concerns the entire labeled core and nominal leximin. It does not cover productive partial-sector coalitions, unequal sector sizes, strictly positive matrices on every allowed edge, ratio-normalized leximin, or a selected proper face of the core. It does not determine the best constant at each finite market size. These restrictions delimit the economic application, rather than follow from the proof. In particular, an arbitrary transportation or extreme-point representation cannot be assumed to preserve all of them.

Companion work. The protected-subsidy companion manuscript permits payments above grand-coalition worth and asks how an external budget restores ideal-based floors. It shares the strict pair-additive market class and individual-core-maxima benchmark, but its sharp subsidy theorem uses a different extremal family. The present paper's graph classification and exact within-core protection result concern no external transfers. Earlier chain–cycle and graph-theorem drafts are superseded presentations of this paper, not additional contributions. The truthful-production companion studies capacity reports and all-profile incentives, assumptions absent here.

Online Resource 1: reproducibility supplement. The self-contained supplement contains rational certificates, exact-solver checks, instructions and source hashes. Tree checks compare cut coordinates and clipping with raw full-team inequalities. Cyclic checks verify matching and cover totals, shortfalls, floors and ideal attainment. The simultaneous checks reconstruct simplex vertices and protection witnesses and test both directions of the simplex description for specified finite parameters. Negative controls reject damaged certificates. Exact coverage is listed in the supplement. These computations corroborate the universal proofs above; they are neither an exhaustive census of markets nor proof-assistant formalization. No verification program is used as a premise of a theorem.

Statements and Declarations

Data and code availability. This theoretical study uses no observed participant or market data. The constructed instances, source code and reproduction instructions are supplied in Online Resource 1. A public archival identifier has not yet been assigned.

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Use of artificial intelligence. OpenAI Codex and Anthropic Claude assisted with mathematical exploration, proof drafting, literature triage, verification code and editorial revision. The supplement distinguishes executed checks, model reviews and unresolved source comparisons. These tools are not authors or human referees. The human author must review the final manuscript and accept responsibility for its claims before submission; that approval remains outstanding for this candidate.

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